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**Auditing the Calibration of
Eligibility Scores
Produced by Large Language
Models
in Patient–Clinical Trial Matching**

*Assessing the Effect of RLHF and Uncovering Mode
Collapse in Open-Source Medical LLMs*

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Abstract

Keywords: large language models, calibration, clinical trial matching, RLHF, medical AI, mode collapse, expected calibration error, conformal prediction

Résumé

Mots-clés : grands modèles de langage, calibration, appariement d’essais cliniques, RLHF, IA médicale, mode collapse, expected calibration error, prédiction conforme

Contents

Acknowledgements	i
Abstract	ii
Résumé	iii
List of Abbreviations	x
1 Introduction	1
1.1 Macro Context: AI in Healthcare	1
1.2 Micro Context: Clinical Trial Matching	1
1.3 Problem Statement	1
1.4 Research Question and Hypotheses	1
1.5 Contributions	1
1.6 Structure of the Thesis	1
2 State of the Art	2
2.1 Patient–Clinical Trial Matching	3
2.1.1 Clinical Context and Need	3
2.1.2 Pre-LLM Approaches	3
2.1.3 LLM-based Approaches	3
2.2 Probabilistic Calibration of Classifiers	3
2.2.1 Definitions and Metrics	3
2.2.2 Clinical Importance of Calibration	3
2.2.3 Recalibration Methods	3
2.3 Calibration of Large Language Models	3
2.3.1 The RLHF Effect on Calibration	3
2.3.2 Calibration of Medical LLMs	3
2.3.3 Identified Gap	3
2.4 Open-Source Medical LLMs	3
2.4.1 The Meditron Family	3
2.4.2 BioMistral and Model Merges	3
2.4.3 Other Open-Source Medical LLMs	3
2.5 Positioning of This Work	3
3 Project Context and Scope	5

3.1	Academic Research Framework	5
3.2	Project Origin and Evolution	5
3.3	Dataset: TrialGPT-Criterion-Annotations	5
3.3.1	Dataset Description	5
3.3.2	Justification of the Single-Dataset Choice	5
3.4	Ethical Considerations	5
3.5	Constraints and Resources	5
3.5.1	Computational Constraints	5
3.5.2	Model Access Constraints	5
3.5.3	Time Constraints	5
3.6	Scope of This Work	5
4	Methodology and Technical Approach	6
4.1	Overview of the Experimental Design	6
4.2	Research Question and Pre-registered Hypotheses	7
4.3	Data Preparation	8
4.3.1	Dataset Composition	8
4.3.2	Label Distribution	9
4.3.3	Textual Characteristics	10
4.3.4	Split for Recalibration Analysis	10
4.3.5	Ethical and Reproducibility Considerations	10
4.4	Model Selection	10
4.4.1	Selection Criteria	11
4.4.2	The Six Selected Models	11
4.4.3	Excluded Models	12
4.5	Prompt Design	13
4.5.1	Rationale for Two Prompt Variants	13
4.5.2	Prompt A: Generic Baseline	13
4.5.3	Prompt B: Adapted with Expert Guidance	14
4.5.4	Adaptations for Base vs. Instruct Models	15
4.6	Scoring Pipeline: Log-Probability Extraction	15
4.6.1	Principle of Log-Probability Scoring	15
4.6.2	Restriction of the Candidate Set by Criterion Type	16
4.6.3	Algorithmic Summary	16
4.6.4	Implementation Notes	17
4.7	Quantization and Infrastructure	17
4.8	Evaluation Metrics	18
4.8.1	Expected Calibration Error (ECE)	18
4.8.2	Brier Score	18

4.8.3	Classification Metrics	19
4.8.4	C-index for Ordinal Validity	19
4.8.5	Weighted Kappa for Expert-Model Agreement	19
4.8.6	Overconfidence and Prediction Diversity	20
4.8.7	Composite Utility Score	20
4.8.8	Reliability Diagrams	21
4.8.9	Bootstrap Confidence Intervals	21
4.9	Statistical Analysis Plan	21
4.9.1	Pre-registration of Hypotheses	21
4.9.2	Permutation Tests	21
4.9.3	Bonferroni Correction	22
4.9.4	Post-hoc Recalibration Methods	22
4.9.5	Conformal Prediction	22
4.10	Reproducibility	23
5	Results and Analysis	24
5.1	Chapter Overview	24
5.2	Pre-LLM Audit and Expert Agreement	24
5.3	Calibration Audit: H1 Supported Across All Models	24
5.4	The Effect of RLHF: H2 Supported (Main Contribution)	25
5.5	The Enriched-Prompt Paradox	25
5.6	Mode Collapse in Medical and RLHF-Aligned LLMs	25
5.7	Performance on Truly Discriminative Cases	25
5.8	Systematic Inclusion vs Exclusion Asymmetry	26
5.9	Model-Specific Evasion Biases	26
5.10	Ordinal Validity (H3) and Selective Classification (H4)	27
5.10.1	H3: Not Reached but Revealing Hierarchy	27
5.10.2	H4: Supported Only on the Mistral Family	27
5.11	Post-hoc Recalibration and Stability	27
5.12	Conformal Prediction and Inter-Model Consensus	27
5.12.1	Conformal Prediction Achieves Target Coverage	27
5.12.2	Inter-Model Consensus	27
5.13	Baselines and Composite Utility Score	27
5.14	Summary of Results	27
6	Conclusion and Future Work	28
6.1	Summary of Findings	28
6.2	Answer to the Research Question	28
6.3	Contributions	28
6.4	Limitations	28

6.4.1	External Validation	28
6.4.2	Model Scale	29
6.4.3	Prompt Space	29
6.5	Future Work	29
6.5.1	External Validation on Independent Benchmarks	29
6.5.2	Multi-Agent Adversarial Debate for Calibration	29
6.5.3	Investigating the Origin of Mode Collapse	29
6.5.4	Extension to Larger and Multilingual Models	29
6.6	Closing Remarks	29
	Bibliography	30
	A Detailed Results Tables	31
	B Full Prompt Specifications	32
B.1	Prompt A — Generic (Baseline)	32
B.2	Prompt B — Adapted with Expert Guidance	32
	C Individual Reliability Diagrams	33
	D Code and Data Availability	34
D.1	Repository Structure	34
D.2	Data Availability	34
D.3	Reproducing the Experiments	34
D.4	Computational Requirements	34
	E Generative AI Usage Declaration	35
	F Detailed Mode Collapse Analysis	37
F.1	Prediction Distribution per Configuration	37
F.2	Simulation: Forcing Decisive Predictions	37
F.3	Composite Utility Score Ranking	37
F.4	Cross-Validation Stability	37

List of Figures

- 4.1 Overview of the experimental design. Six open-source LLMs are evaluated with two prompt variants on 1,015 patient–criterion pairs, producing twelve configurations tested against four pre-registered hypotheses. . . . 7
- 5.1 Reliability diagrams of all 12 configurations (Prompt A left, Prompt B right). All curves fall well below the perfect calibration diagonal. . . . 24
- 5.2 Risk-coverage curves for the 6 Prompt-B configurations. 27
- C.1 Reliability diagrams for the 12 configurations. 33

List of Tables

- 2.1 Comparative positioning of related works and this thesis. 4
- 4.1 The six eligibility labels used in TrialGPT-Criterion-Annotations, organized by criterion type. 9
- 4.2 Expert-label distribution in TrialGPT-Criterion-Annotations. 9
- 4.3 Selected models and their scientific role in the two-by-two experimental matrix. 11
- 4.4 The two-by-two experimental matrix of selected models. 12
- 5.1 Calibration metrics across the 12 configurations. 24
- 5.2 Global vs. non-trivial accuracy across configurations. The performance drop reveals which models rely on the majority class distribution. . . . 26

List of Abbreviations

Acronym	Definition
AUC	Area Under the Curve
CI	Confidence Interval
ECE	Expected Calibration Error
EHR	Electronic Health Record
GPU	Graphics Processing Unit
IC	Intervalle de Confiance (French)
LLM	Large Language Model
LoRA	Low-Rank Adaptation
MC	Mode Collapse
ML4H	Machine Learning for Health
NEI	Not Enough Information
NF4	Normal Float 4-bit (quantization)
NLL	Negative Log-Likelihood
NLP	Natural Language Processing
PPO	Proximal Policy Optimization
QA	Question Answering
RCT	Randomized Controlled Trial
RLHF	Reinforcement Learning from Human Feedback
SFT	Supervised Fine-Tuning
TREC	Text REtrieval Conference

Chapter 1

Introduction

- 1.1 Macro Context: AI in Healthcare
- 1.2 Micro Context: Clinical Trial Matching
- 1.3 Problem Statement
- 1.4 Research Question and Hypotheses
- 1.5 Contributions
- 1.6 Structure of the Thesis

Chapter 2

State of the Art

2.1 Patient–Clinical Trial Matching

2.1.1 Clinical Context and Need

2.1.2 Pre-LLM Approaches

2.1.3 LLM-based Approaches

2.2 Probabilistic Calibration of Classifiers

2.2.1 Definitions and Metrics

2.2.2 Clinical Importance of Calibration

2.2.3 Recalibration Methods

2.3 Calibration of Large Language Models

2.3.1 The RLHF Effect on Calibration

2.3.2 Calibration of Medical LLMs

2.3.3 Identified Gap

2.4 Open-Source Medical LLMs

2.4.1 The Meditron Family

2.4.2 BioMistral and Model Merges

2.4.3 Other Open-Source Medical LLMs

2.5 Positioning of This Work

Table 2.1: Comparative positioning of related works and this thesis.

Work	Domain	Models	RLHF	Calib.	Recalib.
Jin et al. 2024	Trial match- ing	GPT-4	—	—	—
Kadavath 2022	QA	Anthropic	✓	✓	—
Labrak et al. 2024	Med QA	BioMistral	—	✓	—
This work	Trial match- ing	6 LLMs	✓	✓	✓

Chapter 3

Project Context and Scope

3.1 Academic Research Framework

3.2 Project Origin and Evolution

3.3 Dataset: TrialGPT-Criterion-Annotations

3.3.1 Dataset Description

3.3.2 Justification of the Single-Dataset Choice

3.4 Ethical Considerations

3.5 Constraints and Resources

3.5.1 Computational Constraints

3.5.2 Model Access Constraints

3.5.3 Time Constraints

3.6 Scope of This Work

Chapter 4

Methodology and Technical Approach

4.1 Overview of the Experimental Design

This chapter describes the methodology used to audit the calibration of large language models in patient-clinical trial matching. The experimental design follows a pre-registered protocol consistent with the TRIPOD+AI reporting guidelines [1], which recommend explicit hypothesis specification, statistical rigor, and reproducibility for clinical prediction models.

The evaluation pipeline is organized around three principles. First, all predictions are generated on a single held-out benchmark, TrialGPT-Criterion-Annotations [2], which contains 1,015 patient-criterion pairs annotated by three physicians and by GPT-4. Second, model probabilities are extracted directly from the log-probabilities of candidate labels rather than from free-form generation, in order to obtain well-defined probability distributions suitable for calibration analysis. Third, every hypothesis test is defined before the analysis begins, and p-values are corrected for multiple comparisons.

The design forms a 2×2 matrix crossing model type and post-training regime. Along one axis, we compare generalist models (Mistral-7B, Llama-3.1-8B) with medically specialized models (Meditron-7B, BioMistral-7B). Along the other axis, we compare base models (without RLHF) with instruction-tuned models (with RLHF). This crossing produces six models, each of which is evaluated with two prompt variants: a generic minimalist prompt (Prompt A) and an adapted prompt with expert guidance (Prompt B). The full factorial yields twelve configurations, all evaluated on the same 1,015 pairs, for a total of 12,180 predictions.

Figure 4.1 summarizes the pipeline. Section 4.2 states the four pre-registered hypotheses. Section 4.3 describes the dataset. Sections 4.4 and 4.5 justify the model and prompt selection. Section 4.6 details the log-probability scoring pipeline. Section 4.8 defines the calibration and validity metrics. Section 4.9 presents the statistical analysis plan, including the recalibration and conformal prediction procedures.

Figure 4.1: Overview of the experimental design. Six open-source LLMs are evaluated with two prompt variants on 1,015 patient–criterion pairs, producing twelve configurations tested against four pre-registered hypotheses.

4.2 Research Question and Pre-registered Hypotheses

The main research question addressed in this work is stated as follows:

Are the eligibility scores produced by large language models in patient–clinical trial matching probabilistically reliable, and does reinforcement learning from human feedback affect this reliability?

This question is decomposed into four falsifiable hypotheses, each associated with a numerical rejection threshold defined prior to the experimental phase. Pre-registration of hypotheses is a standard safeguard against confirmation bias in empirical work [1] and provides a clear criterion for evaluating each finding.

Hypothesis H1 tests whether current LLM pipelines are systematically miscalibrated. The rejection threshold is derived from Van Calster et al. [3], who characterize an Expected Calibration Error (ECE) below 0.05 as evidence of a well-calibrated predictive system in clinical contexts.

Hypothesis 1 (H1: Poor calibration of current LLM pipelines). *The Expected Calibration Error of an LLM-based eligibility scoring pipeline exceeds the well-calibrated threshold of 0.05:*

$$ECE(\text{pipeline}) > 0.05.$$

Hypothesis H2 extends the finding of Kadavath et al. [4], who observed that RLHF post-training degrades calibration on general question-answering tasks, to the specific domain of clinical trial matching. It is the central original contribution of this thesis. The comparison uses a permutation test on the difference in ECE between paired Base and Instruct versions of the same model family, with Bonferroni correction across families and prompt variants.

Hypothesis 2 (H2: RLHF degrades calibration). *Within a given model family, the RLHF-aligned Instruct variant exhibits worse calibration than its Base counterpart:*

$$ECE(\text{Instruct}) > ECE(\text{Base}).$$

Hypothesis H3 tests whether, independently of absolute calibration, the model’s scores

respect the ordinal structure of the eligibility labels. Ordinal validity is measured by the concordance index (C-index), and the threshold of 0.70 corresponds to conventional discrimination criteria used in clinical prediction modeling [3].

Hypothesis 3 (H3: Ordinal validity). *The C-index between the LLM ordinal score and the expert ordinal label exceeds the standard discrimination threshold:*

$$\text{C-index} > 0.70.$$

Hypothesis H4 addresses the practical value of confidence-based abstention. If confidence scores are meaningful, restricting predictions to the most confident subset should yield higher classification quality. The threshold of +0.02 on macro-F1 gain is chosen to reflect a clinically noticeable improvement without being trivially achievable.

Hypothesis 4 (H4: Selective classification improves quality). *Restricting predictions to the top 80% most confident cases increases macro-F1 by at least 0.02:*

$$F_1(\text{coverage} = 80\%) > F_1(\text{coverage} = 100\%) + 0.02.$$

Together, these four hypotheses probe distinct aspects of probabilistic reliability: absolute calibration (H1), the effect of RLHF (H2), ordinal consistency (H3), and practical usability under abstention (H4). All rejection criteria and statistical tests were fixed prior to running the experiments and are reported without modification in Chapter 5.

4.3 Data Preparation

The experimental evaluation is conducted on the TrialGPT-Criterion-Annotations dataset [2], released alongside the TrialGPT system by Jin et al. To our knowledge, it is the only publicly available benchmark that provides multi-level expert annotations at the level of individual eligibility criteria, with an explicit distinction between inclusion and exclusion criteria. These properties are essential for a fine-grained calibration analysis: aggregating annotations at the trial level, as in TREC Clinical Trials or SIGIR benchmarks, would preclude the per-criterion probability estimation required by the present protocol.

4.3.1 Dataset Composition

The dataset comprises 1,015 patient–criterion pairs. Each pair associates a synthetic patient note with a single eligibility criterion drawn from a real clinical trial. Two independent annotations are provided per pair: a consensus label from three physicians,

and a label produced by GPT-4. Both annotations use the same six-level ordinal scale, described in Table 4.1.

Table 4.1: The six eligibility labels used in TrialGPT-Criterion-Annotations, organized by criterion type.

Label	Criterion type	Semantics
included	inclusion	The patient satisfies the inclusion criterion.
not included	inclusion	The patient does not satisfy the criterion.
excluded	exclusion	The patient has the excluding condition.
not excluded	exclusion	The patient does not have the excluding condition (favorable).
not enough information	both	The note does not permit a decision.
not applicable	both	The criterion does not apply to this patient.

Two of the six labels function as evasion or uncertainty markers rather than decisive clinical judgments (“not enough information” and “not applicable”). We deliberately keep them in the label space to preserve the semantics of the original benchmark, while treating their behavior as an object of study in its own right (see Sections 5.9 and 5.6).

4.3.2 Label Distribution

The distribution of expert annotations is markedly imbalanced (Table 4.2). “Not excluded” dominates at 47.7%, reflecting the fact that most patients do not present the specific medical conditions targeted by exclusion criteria. “Not enough information” accounts for a further 28.9%. The remaining 23.4% of pairs, distributed across four labels, represent the cases requiring a truly discriminative clinical judgment. This asymmetry is critical for the interpretation of results in Chapter 5: aggregate metrics computed over the full 1,015 pairs can be dominated by the two majority classes and may mask complete failure on the discriminative subset.

Table 4.2: Expert-label distribution in TrialGPT-Criterion-Annotations.

Label	Count	Proportion
not excluded	484	47.7%
not enough information	293	28.9%
included	74	7.3%
excluded	55	5.4%
not applicable	55	5.4%
not included	54	5.3%
Total	1,015	100.0%

4.3.3 Textual Characteristics

Patient notes are short and structured: their length ranges from 232 to 1,447 characters with a median of 579 characters. This compactness is important for two reasons. First, it means that no aggressive truncation is required to fit the notes within LLM context windows, avoiding an additional source of variability. Second, short notes place a heavier reasoning burden on the model: information is often implicit, and the model must infer unstated conditions from the surrounding clinical context. Criterion texts are also short, with a median length of 58 characters, ranging from declarative statements (*Age 18 to 75 years*) to more elaborate clauses involving numeric thresholds or drug interactions.

4.3.4 Split for Recalibration Analysis

The full dataset is used for the primary calibration audit (Chapter 5, Section 5.3) since the models are queried without any prior fitting on this data. For the post-hoc recalibration analysis (Section 5.11), we adopt a 70/30 random split stratified by criterion type. The 70% subset is used to fit the isotonic regression, the 30% subset to evaluate the recalibrated ECE. Stability of the recalibration gain is further verified through five-fold cross-validation using the same split ratio.

4.3.5 Ethical and Reproducibility Considerations

The dataset contains no real patient data: all notes are synthetic, constructed by the original authors to match the semantics of real recruitment scenarios without disclosing personal health information. No institutional review board approval was therefore required for the present work. The dataset is publicly available under a CC BY 4.0 license, and all prediction files produced in this study are archived in the project repository (see Appendix D) to support independent reproduction of the analyses.

4.4 Model Selection

The choice of models under evaluation is the central design decision of this work. Rather than surveying a broad catalogue of open-source LLMs, we prioritize a small selection organized as a controlled experimental matrix. This design enables clean isolation of the effects under study — particularly the effect of RLHF post-training — while remaining feasible on the limited computational resources available.

4.4.1 Selection Criteria

Three criteria guide the selection of models. First, the model must be available in both a base and an instruction-tuned variant with a common pretraining, since testing hypothesis H2 requires paired comparisons within the same model family. Second, the selection must span the two-by-two matrix of interest: generalist versus medically specialized pretraining, crossed with the presence or absence of RLHF post-training. Third, the model must fit into 16 GB of GPU memory when quantized to 4 bits, which is the memory budget of a free-tier T4 GPU on Kaggle. This constraint effectively caps the parameter count at 7–9 billion.

These three criteria eliminate several candidates that would otherwise be attractive. Models larger than 9 B parameters (Meditron-70B, Med42-v2-70B, Llama-3.3-70B) exceed the memory budget. Model families without a clean base–instruct pairing (Med42-v2 releases only the instruction-tuned variant) preclude the H2 comparison. Model families released without a medical counterpart (Gemma, DeepSeek) do not add information for our two-by-two matrix.

4.4.2 The Six Selected Models

The six selected models, together with their scientific role in the experimental matrix, are summarized in Table 4.3.

Table 4.3: Selected models and their scientific role in the two-by-two experimental matrix.

Model	Type	Scientific Role
MISTRAL-7B-v0.3	Generalist, no RLHF	H2 control
MISTRAL-7B-INSTRUCT-v0.3	Generalist, with RLHF	H2 test
LLAMA-3.1-8B	Generalist, no RLHF	H2 replication
LLAMA-3.1-8B-INSTRUCT	Generalist, with RLHF	H2 replication
MEDITRON-7B	Medical, no RLHF	Pure specialization effect
BIOMISTRAL-7B	Medical, inherited RLHF	Validation vs. Labrak et al. (2024)

MISTRAL-7B-v0.3 and its instruction-tuned counterpart MISTRAL-7B-INSTRUCT-v0.3 form the primary base–instruct pair for testing H2. The two models share the same pretraining corpus and architecture; the Instruct variant differs only through subsequent supervised fine-tuning and RLHF. This clean isolation of the RLHF stage makes them the reference comparison for our central hypothesis.

LLAMA-3.1-8B and LLAMA-3.1-8B-INSTRUCT play the same role for a second, independent model family. Testing H2 on two families rather than one rules out the possibility that any observed effect is specific to a single training pipeline. Llama-3.1

also brings a different tokenizer, a larger context window, and a different alignment methodology, which strengthens the generality of the conclusions.

MEDITRON-7B [5] is a foundation model pretrained on 46,000 clinical practice guidelines and a large corpus of PubMed abstracts, starting from Llama-2 weights. Crucially, no official instruction-tuned variant of Meditron-7B has been released; the base model is the only publicly available version. Meditron therefore isolates the effect of medical pretraining without any confound with RLHF, providing a direct test of whether medical specialization improves calibration in the absence of instruction alignment.

BIOMISTRAL-7B [6] occupies the opposite corner of the two-by-two matrix. It is derived from Mistral-7B-Instruct-v0.1 through continued pretraining on PubMed Central Open Access texts. It therefore inherits the RLHF alignment of its Mistral-Instruct parent while adding a medical specialization stage. BioMistral is also the only open-source medical LLM with published ECE values, which provides an external validation point for our measurement methodology.

Together, these six models populate the two-by-two matrix completely (Table 4.4). Each of the three scientific questions associated with the matrix — the effect of RLHF, the effect of medical specialization, and their interaction — can be tested through paired comparisons within this design.

Table 4.4: The two-by-two experimental matrix of selected models.

	Without RLHF	With RLHF
Generalist	MISTRAL-7B	MISTRAL-7B-INSTRUCT
	LLAMA-3.1-8B	LLAMA-3.1-8B-INSTRUCT
Medical	MEDITRON-7B	BIOMISTRAL-7B

4.4.3 Excluded Models

Several candidate models were considered and deliberately excluded from the final selection. QWEN2.5-7B and DEEPSEEK-V2-LITE were excluded as redundant additions to the generalist axis: they would not provide new evidence for H2 beyond what Mistral and Llama already establish. GEMMA-2-9B was excluded for the same reason, and additionally because its license restricts commercial redistribution of derivative model outputs, complicating reproducibility.

Among medical LLMs, MED42-V2 [7] and OPENBIOLLM-8B were excluded because their base versions are not publicly released, precluding a within-family H2 comparison. HUATUOGPT-O1 [8] was excluded because its training corpus is primarily Chinese, which introduces a confound with the English-only nature of the TrialGPT-Criterion

benchmark. Larger medical models (MEDITRON-70B, MED42-v2-70B, ALOE-70B) all exceed the memory budget of a free-tier T4 GPU, even at 4-bit quantization.

The rationale for this restrictive selection is that a small number of well-matched models tested on the same protocol yields more interpretable conclusions than a broader benchmark that would necessarily mix confounding factors. Extending the analysis to additional families is a natural direction for future work, discussed in Chapter 6.

4.5 Prompt Design

The prompt used to elicit predictions from a large language model is not a neutral component of an evaluation pipeline: it is a design choice that directly affects the model’s outputs. Any calibration audit that fixes a single prompt inevitably conflates two effects — the properties of the model and the properties of the prompt — which cannot be separated post hoc. To address this ambiguity, we evaluate each of the six models under two distinct prompt variants, treating the prompt as an experimental variable rather than a fixed convention.

4.5.1 Rationale for Two Prompt Variants

The two prompts are designed to represent the two extremes of a spectrum of elicitation strategies commonly encountered in practice. The first (Prompt A) is intentionally minimal: it presents the classification task and the label vocabulary without any explanation, in the manner of a naive user querying the model. The second (Prompt B) is enriched with definitions, examples, and explicit reasoning rules, in the manner of an expert operator who invests time in prompt engineering. By evaluating each model with both prompts, we obtain four independent measurement points per model family for hypothesis H2 (Base–Instruct comparison at Prompt A, Base–Instruct comparison at Prompt B), doubling the statistical power of the RLHF test.

This design also enables three additional analyses beyond the pre-registered hypotheses. First, the effect of prompt design itself becomes measurable within each model. Second, the interaction between RLHF and prompt design becomes testable: some models may be more sensitive to prompt enrichment than others. Third, the design places our results in a stronger position with respect to external validity: a finding replicated under two different prompts is more robust than a finding measured under a single one.

4.5.2 Prompt A: Generic Baseline

Prompt A is a minimalist prompt containing only the classification instruction, the input note, the criterion text, and the list of candidate labels. It does not define the

labels, does not provide examples, and does not include reasoning rules. Its purpose is to establish a baseline that corresponds to how a first-time user would query the model, and to provide a reference point against which enriched prompts can be compared.

The complete text of Prompt A is given in Appendix B. Its essential structure is:

Classify whether this patient meets the eligibility criterion.

Patient: [note]

Criterion ([type]): [criterion]

Choose exactly one label from: included, not included, excluded, not excluded, not enough information, not applicable.

Answer:

Prompt A is applied identically to all six models. For instruction-tuned models, the prompt is wrapped in the model’s native chat template. For base models, the prompt is presented as a plain-text continuation task without special tokens.

4.5.3 Prompt B: Adapted with Expert Guidance

Prompt B is an adapted prompt containing definitions of the six labels, one worked example per label, and a critical inference rule for exclusion criteria that addresses a subtle failure mode observed in preliminary experiments. The full text is reproduced in Appendix B.

Three design choices distinguish Prompt B from Prompt A. First, each of the six labels is defined with an explicit semantic description. For example, “not included” is characterized as “The patient note contains information showing the patient does NOT meet this inclusion criterion,” which distinguishes it from “not enough information” (where the note is silent). Second, a one-sentence example is provided for each label, drawn from realistic clinical situations. Third, a specific inference rule is stated for exclusion criteria:

If the criterion names a specific disease, condition, or medical history, and the note does not mention it, the default is “not excluded” (the patient most likely does not have it). Reserve “not enough information” only for criteria requiring a specific lab value or measurement that is truly missing.

This rule directly targets the tendency of language models to produce “not enough information” for any exclusion criterion whose target condition is not explicitly discussed in the patient note — a pattern that mechanically converts the absence of a mention into an expression of epistemic uncertainty, when clinically it should be treated as evidence of absence.

4.5.4 Adaptations for Base vs. Instruct Models

Base models and instruction-tuned models expect prompts in different formats, and applying the same prompt verbatim to both would create an artificial handicap for the base variant. To ensure a fair comparison, we adapt the surface form of each prompt to the target model class while preserving its semantic content.

For instruction-tuned models, prompts are formatted using the model’s native chat template, which delimits user and assistant roles with model-specific special tokens ([INST] for Mistral, <|start_header_id|> for Llama-3). This is the format for which these models were aligned and where they behave as expected.

For base models, no chat template is applied. Instead, the prompt includes one or more few-shot examples that establish the desired input-output pattern by demonstration. Prompt A for base models includes a single example; Prompt B includes three examples covering distinct label types. This asymmetry reflects a genuine limitation of base models — they do not natively follow instructions — and treating it as an evaluation artifact would misrepresent how these models are actually used in practice. The introduction of few-shot examples is nevertheless a potential confounder for cross-family comparisons, and we discuss it explicitly in Chapter 6.

The complete text of all four prompt variants (Prompt A / Prompt B, Instruct / Base) is provided in Appendix B to support independent reproduction of the pipeline.

4.6 Scoring Pipeline: Log-Probability Extraction

Standard practice for querying a large language model on a classification task is to sample a textual output and post-process it to recover a label. This approach has two limitations for a calibration audit. First, the sampled output is a point estimate: it discards the underlying probability distribution over candidate labels, which is precisely the object of interest for calibration analysis. Second, generation is stochastic, introducing an additional source of variability that would complicate the reproducibility of downstream statistics. We therefore adopt a log-probability scoring approach that directly reads out the model’s probability distribution over the candidate labels, without any generation step.

4.6.1 Principle of Log-Probability Scoring

For a given patient–criterion pair, we first construct the prompt as described in Section 4.5. Rather than sampling a continuation, we consider each candidate label as a hypothetical completion of the prompt. For each label ℓ , we compute the mean

log-probability of its token sequence conditional on the prompt:

$$s_\ell = \frac{1}{|\ell|} \sum_{t=1}^{|\ell|} \log p(\ell_t \mid \text{prompt}, \ell_{<t}) \quad (4.1)$$

where ℓ_t is the t -th token of the label and $|\ell|$ is the total number of tokens. Length normalization by $|\ell|$ is necessary because labels have unequal token lengths in the tokenizer: without it, longer labels such as “not enough information” would be systematically penalized against shorter ones such as “included”.

The resulting scores $\{s_\ell\}$ are then normalized through a softmax to obtain a proper probability distribution π over the candidate labels:

$$\pi_\ell = \frac{\exp(s_\ell)}{\sum_{\ell' \in \mathcal{L}} \exp(s_{\ell'})} \quad (4.2)$$

The predicted label $\hat{y}$ is the argmax of π , and the associated confidence $\hat{p}$ is the maximum probability. Crucially, the full distribution π is retained for downstream analyses, including recalibration, conformal prediction, and the simulation of constrained label spaces (Section 5.6).

4.6.2 Restriction of the Candidate Set by Criterion Type

The six labels described in Section 4.3 split into two groups by criterion type. For inclusion criteria, the semantically valid labels are *included*, *not included*, *not enough information*, and *not applicable*. For exclusion criteria, they are *excluded*, *not excluded*, *not enough information*, and *not applicable*. In both cases, four labels are candidates and two would be semantically nonsensical (e.g., predicting “included” for an exclusion criterion). We therefore restrict the softmax in Equation 4.2 to the four candidates appropriate to the criterion type at hand. This restriction preserves the ordinal structure of the task and prevents the model from being penalized for probability mass placed on labels that could not have been correct in the first place.

4.6.3 Algorithmic Summary

Algorithm 1 summarizes the full scoring procedure applied to each patient–criterion pair.

Because no sampling is involved, the procedure is fully deterministic: running the same model on the same input twice yields identical scores. This determinism is important for the reproducibility of the entire downstream analysis, including the bootstrap confidence intervals and permutation tests described in Section 4.9.

Algorithm 1 Log-probability scoring for eligibility label prediction

Require: Patient note n , criterion c , criterion type t **Ensure:** Predicted label $\hat{y}$, confidence $\hat{p}$, distribution π

```

1: Build prompt from  $n, c, t$ 
2:  $\mathcal{L} \leftarrow$  candidate labels according to  $t$ 
3: for all  $\ell \in \mathcal{L}$  do
4:    $s_\ell \leftarrow$  mean log-probability of tokens of  $\ell$ 
5: end for
6:  $\pi \leftarrow \text{softmax}(s)$ 
7:  $\hat{y} \leftarrow \arg \max_\ell \pi_\ell$ 
8:  $\hat{p} \leftarrow \pi_{\hat{y}}$ 
9: return  $\hat{y}, \hat{p}, \pi$ 

```

4.6.4 Implementation Notes

The scoring loop is implemented using the standard causal language modeling interface of the HuggingFace `transformers` library. For a given prompt, we tokenize the concatenation of the prompt and each candidate label separately, run a single forward pass per concatenation, and read out the token log-probabilities from the model’s logits after applying a log-softmax over the vocabulary. This approach requires four forward passes per patient–criterion pair (one per candidate label), which is approximately twice as expensive as free-form generation of a short answer. In practice, on a T4 GPU with 4-bit quantization, the four passes complete in approximately three seconds per pair for a 7-billion-parameter model, and the full evaluation of 1,015 pairs takes fifty to seventy minutes depending on the model.

4.7 Quantization and Infrastructure

Running 7-to-9-billion-parameter models on a free-tier T4 GPU (16 GB of memory) requires reducing the memory footprint of the model weights. We apply 4-bit NF4 quantization **dettmers2023qlora** through the `bitsandbytes` library, which represents each weight as a 4-bit value drawn from a distribution matched to the empirical distribution of neural network weights. This reduces the memory required to store a 7-billion-parameter model from approximately 14 GB (in half-precision) to approximately 4 GB, leaving sufficient headroom for the key-value cache during inference.

Prior work has shown that 4-bit NF4 quantization introduces negligible degradation on downstream evaluation benchmarks compared to full precision **dettmers2023qlora**, including on tasks that involve long-form generation or fine-grained probability estimation. We rely on this evidence to justify the use of NF4 quantization for our calibration audit, while acknowledging that a small residual effect of quantization on absolute

ECE values cannot be entirely ruled out. Since Hypothesis H2 and the associated Bonferroni-corrected permutation tests compare *differences* in ECE between paired variants evaluated under the same quantization regime, any systematic bias introduced by quantization cancels out and does not affect the conclusions.

All experiments were run on Kaggle notebooks with a single Tesla T4 GPU. The compute quota (30 GPU-hours per week on the free tier) was sufficient to complete the twelve runs and all downstream analyses within the experimental phase of the project.

4.8 Evaluation Metrics

The evaluation combines calibration metrics, classification metrics, ordinal validity metrics, and behavioral metrics. Each captures a distinct aspect of probabilistic reliability, and no single metric is sufficient on its own. This section defines every metric used in Chapter 5.

4.8.1 Expected Calibration Error (ECE)

The Expected Calibration Error, introduced by Naeini et al. and popularized by Guo et al. [9], measures the average absolute gap between predicted confidence and observed accuracy across confidence bins:

$$\text{ECE} = \sum_{k=1}^K \frac{|B_k|}{n} |\text{acc}(B_k) - \text{conf}(B_k)| \quad (4.3)$$

where predictions are partitioned into K equal-width bins $B_1, \dots, B_K$ based on their confidence, n is the total number of predictions, $\text{acc}(B_k)$ is the empirical accuracy of predictions in bin B_k , and $\text{conf}(B_k)$ is the average confidence of predictions in the same bin. A perfectly calibrated model satisfies $\text{acc}(B_k) = \text{conf}(B_k)$ for all bins, yielding an ECE of zero. We use $K = 10$ equal-width bins on $[0, 1]$.

4.8.2 Brier Score

The Brier score **brier1950** measures the mean squared error between predicted probability and the true binary outcome (correct or incorrect):

$$\text{Brier} = \frac{1}{n} \sum_{i=1}^n (p_i - y_i)^2 \quad (4.4)$$

where p_i is the predicted confidence and $y_i \in \{0, 1\}$ indicates whether the prediction was correct. Unlike ECE, which measures calibration alone, the Brier score jointly

reflects calibration and sharpness: a model that is both accurate and confident achieves a lower Brier score than one that is only well calibrated but uninformative.

4.8.3 Classification Metrics

We report accuracy as the proportion of predictions matching the expert label, and macro-F1 as the unweighted mean of per-class F1 scores. Macro-F1 is critical in the presence of the severe class imbalance documented in Section 4.3.2: a model that always predicts the majority class achieves an accuracy of 47.7% but a macro-F1 close to zero on the minority classes. This distinction becomes central in Chapter 5 when comparing model behavior across cases of varying clinical difficulty.

4.8.4 C-index for Ordinal Validity

Because the eligibility labels form an ordinal scale, we complement calibration and classification metrics with the concordance index (C-index), which measures the probability that, for a randomly chosen pair of cases with different expert labels, the model correctly ranks them:

$$\text{C-index} = \frac{|\{(i, j) : y_i < y_j \text{ and } s_i < s_j\}|}{|\{(i, j) : y_i \neq y_j\}|} \quad (4.5)$$

where y_i, y_j are ordinal expert labels of cases i and j , and s_i, s_j are the model’s scalar scores obtained by mapping the predicted label and confidence onto the ordinal scale. A C-index of 1 indicates perfect ranking, 0.5 indicates random ranking, and the conventional threshold of 0.70 is used in clinical prediction modeling as evidence of acceptable discrimination [3].

4.8.5 Weighted Kappa for Expert-Model Agreement

To measure the agreement between physician annotations and GPT-4 annotations in the pre-LLM audit (Section 5.2), we use Cohen’s weighted kappa with quadratic weights **cohen1968**. Weighted kappa extends the standard kappa coefficient by penalizing disagreements proportionally to their ordinal distance: a confusion between adjacent labels (e.g., “included” and “not enough information”) counts less than a confusion between distant labels (e.g., “included” and “excluded”). We report values in the range $[-1, 1]$, with 1 indicating perfect agreement, 0 indicating chance agreement, and negative values indicating systematic disagreement.

4.8.6 Overconfidence and Prediction Diversity

Beyond calibration and classification, two behavioral metrics quantify how a model uses its probability distribution.

The mean overconfidence is defined as the difference between the average confidence and the average accuracy:

$$\text{overconfidence} = \bar{p} - \text{acc} \quad (4.6)$$

where $\bar{p}$ is the mean predicted confidence across the evaluation set. A positive value indicates that the model tends to overstate its certainty, a negative value that it understates it. Unlike ECE, which is a magnitude, overconfidence is a signed quantity that reveals the direction of the miscalibration.

Prediction diversity is measured as the normalized Shannon entropy of the empirical label distribution:

$$\text{diversity} = \frac{-\sum_{\ell \in \mathcal{L}} f_{\ell} \log_2 f_{\ell}}{\log_2 |\mathcal{L}|} \quad (4.7)$$

where f_{ℓ} is the empirical frequency of label ℓ in the model's predictions and $\mathcal{L}$ is the label vocabulary. A diversity of 0 indicates that the model predicts a single label; a diversity of 1 indicates a uniform distribution over all labels. This metric is central to detecting the mode collapse behavior documented in Section 5.6, which the calibration and classification metrics can mask.

4.8.7 Composite Utility Score

To provide a single rank-ordering of the twelve configurations that reflects clinical usability, we propose a composite utility score combining calibration quality, discrimination, and prediction diversity:

$$\text{utility} = F_1^{\text{macro}} \times (1 - \text{ECE}) \times \text{diversity} \quad (4.8)$$

The three factors are multiplicative rather than additive so that a failure on any single dimension zeroes out the score. A model with excellent calibration but complete mode collapse (diversity = 0) receives a utility score of zero, correctly capturing its lack of practical value. This composite score complements rather than replaces the individual metrics, which remain the primary object of analysis.

4.8.8 Reliability Diagrams

Reliability diagrams complement ECE by visualizing the calibration curve directly. For each of the K bins used in the ECE computation, we plot the average confidence on the horizontal axis and the observed accuracy on the vertical axis. A perfectly calibrated model produces points that lie on the diagonal $y = x$. Points below the diagonal indicate overconfidence; points above indicate underconfidence.

4.8.9 Bootstrap Confidence Intervals

For every point estimate of ECE, Brier score, and C-index, we report a 95% confidence interval obtained by 1,000 bootstrap resamples of the prediction set. Bootstrap confidence intervals do not assume any specific distributional form and are appropriate for the non-Gaussian statistics under study. They also provide a natural way to assess whether a given model’s ECE genuinely exceeds the 0.05 threshold of Hypothesis H1, or whether the excess is within sampling noise.

4.9 Statistical Analysis Plan

The four pre-registered hypotheses described in Section 4.2 are tested according to a fixed analysis plan. This section describes the tests, the correction for multiple comparisons, and the additional analyses (post-hoc recalibration and conformal prediction) that complement the primary hypotheses.

4.9.1 Pre-registration of Hypotheses

All four rejection criteria (H1: $ECE > 0.05$; H2: $ECE(\text{Instruct}) > ECE(\text{Base})$; H3: $C\text{-index} > 0.70$; H4: $F_1(80\%) > F_1(100\%) + 0.02$) were fixed before the twelve model runs were executed. This pre-registration protects the analysis against post-hoc adjustment of the criteria to fit the observed results and follows the recommendations of the TRIPOD+AI reporting guidelines [1]. Results are reported without modification of the rejection criteria, and any findings not covered by these four hypotheses are reported separately as original observations (see Chapter 5).

4.9.2 Permutation Tests

Hypothesis H2 compares the ECE of paired Base and Instruct variants of the same model family under the same prompt. Rather than assume any particular distributional form for the ECE difference, we use a permutation test: for 1,000 random permutations of the model labels (Base vs. Instruct) within each pair, we recompute the ECE difference

and count the fraction of permutations in which the observed difference is equalled or exceeded. This fraction is the one-sided p-value for the null hypothesis that the two variants have the same underlying calibration. The one-sided formulation is appropriate because Hypothesis H2 is directional: we test whether Instruct is worse than Base, not merely different from it.

4.9.3 Bonferroni Correction

Hypothesis H2 is tested four times (Mistral–Prompt A, Mistral–Prompt B, Llama–Prompt A, Llama–Prompt B) and further comparisons for medical specialization contribute additional tests. To control the family-wise error rate across the eight planned comparisons, we apply a Bonferroni correction with $\alpha_{\text{corrected}} = 0.05/8 = 0.00625$. This correction is conservative but appropriate given the small number of comparisons and the desire for a stringent test of the central hypothesis.

4.9.4 Post-hoc Recalibration Methods

Beyond the primary calibration audit, we evaluate three post-hoc recalibration methods that could correct miscalibration without retraining the model. Temperature scaling [9] applies a single learned scalar T to the logits before the softmax, effectively softening (or sharpening) the confidence distribution. Isotonic regression fits a monotone piecewise-constant function mapping uncorrected confidences to corrected ones; being non-parametric, it can correct arbitrary shapes of miscalibration but requires more data than temperature scaling. Platt scaling fits a logistic function of the uncorrected confidence, offering a compromise between the flexibility of isotonic regression and the parsimony of temperature scaling.

Each method is fitted on the 70% training subset and evaluated on the 30% held-out subset (see Section 4.3.4). To assess whether the recalibration gains are stable across different data splits, we additionally perform a 5-fold cross-validation, reporting the mean and standard deviation of the ECE reduction over the five folds.

4.9.5 Conformal Prediction

To complement point-estimate calibration, we apply split conformal prediction **vovk2005**, a distribution-free framework that produces prediction sets with a guaranteed marginal coverage rate. Given a target coverage level $1 - \alpha$ (we use $\alpha = 0.10$, i.e., 90% coverage), the framework calibrates a threshold on the non-conformity scores of a held-out calibration set and returns, for each test case, the set of labels whose non-conformity is below that threshold. The size of the prediction set is an informative quantity: large sets indicate high aleatoric uncertainty, while small sets indicate high model confidence.

We report both empirical coverage (which should match the target 90%) and average set size.

4.10 Reproducibility

All code, prompts, and prediction files produced in this study are archived in the project repository listed in Appendix D. The code is organized as a single Kaggle notebook that reproduces the twelve model runs from scratch, together with a set of Python scripts that perform the downstream analyses (calibration audit, hypothesis tests, recalibration, and conformal prediction) from the archived prediction files. Random seeds are fixed throughout the pipeline: the 4-bit quantization is deterministic, the log-probability scoring is deterministic by construction (Section 4.6), and the bootstrap and permutation tests use a fixed seed to ensure that statistical results are exactly reproducible.

The complete list of hardware and software versions used in the experiments, together with the exact commit hashes of the notebook and the analysis scripts, is provided in Appendix D. Any independent researcher with access to a T4 GPU (or equivalent) can reproduce the twelve model runs in approximately ten to fifteen hours of GPU time, and the downstream analyses in a few additional minutes on CPU.

Chapter 5

Results and Analysis

5.1 Chapter Overview

5.2 Pre-LLM Audit and Expert Agreement

Finding 1 (F1: GPT-4’s Selective Evasion Bias). *GPT-4 over-uses the “not applicable” label (ratio 2.16 vs experts) but uses “not enough information” with a ratio of 0.99. This asymmetric pattern reveals a selective evasion, unreported in the original TrialGPT paper.*

5.3 Calibration Audit: H1 Supported Across All Models

Table 5.1: Calibration metrics across the 12 configurations.

Configuration	ECE	95% CI	Brier
Mistral-Base A	0.133	[0.104, 0.163]	0.258
Mistral-Instruct A	0.322	[0.297, 0.350]	0.311
Mistral-Base B	0.189	[0.161, 0.218]	0.269
Mistral-Instruct B	0.253	[0.229, 0.287]	0.353
Llama-Base A	0.124	[0.100, 0.148]	0.187
Llama-Instruct A	0.510	[0.483, 0.535]	0.456
Llama-Base B	0.151	[0.123, 0.177]	0.225
Llama-Instruct B	0.602	[0.575, 0.628]	0.558
Meditron A	0.182	[0.153, 0.212]	0.263
Meditron B	0.231	[0.204, 0.259]	0.283
BioMistral A	0.155	[0.126, 0.182]	0.234
BioMistral B	0.194	[0.166, 0.222]	0.244

Figure 5.1: Reliability diagrams of all 12 configurations (Prompt A left, Prompt B right). All curves fall well below the perfect calibration diagonal.

5.4 The Effect of RLHF: H2 Supported (Main Contribution)

Finding 2 (F2: RLHF Systematically Degrades Calibration). *Across two independent model families (Mistral-7B and Llama-3.1-8B) and two independent prompt designs, RLHF-aligned Instruct versions exhibit significantly worse calibration than their Base counterparts (all $p < 0.003$ with Bonferroni correction, ΔECE ranging from $+0.064$ to $+0.451$).*

5.5 The Enriched-Prompt Paradox

Finding 3 (F3: The Enriched-Prompt Paradox). *Contrary to the intuition that expert prompt design improves calibration, prompts enriched with definitions, examples, and exclusion rules degrade calibration in 5 of 6 models. Only the strongly RLHF-aligned Mistral-Instruct benefits.*

5.6 Mode Collapse in Medical and RLHF-Aligned LLMs

Finding 4 (F7: Mode Collapse in 5 of 12 Configurations). *Five open-source configurations — both BioMistral variants, all Llama-Instruct variants, and Llama-Base with prompt B — predict “not enough information” in more than 97% of cases. Their apparently reasonable ECE (0.15–0.60) reflects proximity to a trivial NEI baseline rather than genuine probabilistic quality.*

Finding 5 (F8: Mode Collapse Is Structural, Not Superficial). *When forced to choose between the two decisive labels of each criterion type by removing evasion candidates from the softmax, collapsed models do not recover reasoning ability. Instead, collapse migrates from “not enough information” to the majority label per criterion type (“not excluded” for exclusion criteria, or “excluded” for one Llama-Base variant). Evasion behaviors mask absence of clinical reasoning rather than strategic uncertainty communication.*

5.7 Performance on Truly Discriminative Cases

Finding 6 (F9: Global Accuracy Hides Complete Failure on Discriminative Cases). *On the 238 truly discriminative cases (23.4% of the dataset), Meditron achieves 0% accuracy despite its 64.8% global accuracy; BioMistral achieves 0.0% and 0.4% accuracy for its two prompts. The apparent superiority of medical LLMs disappears on the*

cases requiring genuine clinical decisions. Only Mistral-Instruct with prompt A reaches acceptable clinical performance (71.9% accuracy, ECE 0.054).

Table 5.2: Global vs. non-trivial accuracy across configurations. The performance drop reveals which models rely on the majority class distribution.

Configuration	Global acc.	Non-trivial acc.	Drop
Meditron A	0.648	0.000	−0.847
Meditron B	0.630	0.000	−0.824
BioMistral B	0.289	0.000	−0.377
BioMistral A	0.296	0.004	−0.381
Llama-Instruct B	0.292	0.013	−0.365
Llama-Base B	0.308	0.013	−0.386
Llama-Instruct A	0.290	0.021	−0.351
Llama-Base A	0.230	0.265	+0.046
Mistral-Base B	0.586	0.416	−0.222
Mistral-Instruct B	0.480	0.500	+0.026
Mistral-Base A	0.500	0.613	+0.149
Mistral-Instruct A	0.308	0.719	+0.536

5.8 Systematic Inclusion vs Exclusion Asymmetry

Finding 7 (F4: Inclusion vs Exclusion Asymmetry). *LLM calibration on exclusion criteria is 2–3 times worse than on inclusion criteria across all generalist models. MEDITRON is the sole exception, showing the reverse pattern — likely due to its clinical guidelines training data.*

5.9 Model-Specific Evasion Biases

Finding 8 (F5: Divergent Evasion Strategies Across LLMs). *Whereas GPT-4 evades via the “not applicable” label (ratio 2.16), open- source LLMs consistently evade via “not enough information” (ratio 3.3 to 3.5 for BioMistral and Llama-Instruct). Evasion is systematic but channel-specific.*

Figure 5.2: Risk-coverage curves for the 6 Prompt-B configurations.

5.10 Ordinal Validity (H3) and Selective Classification (H4)

5.10.1 H3: Not Reached but Revealing Hierarchy

5.10.2 H4: Supported Only on the Mistral Family

5.11 Post-hoc Recalibration and Stability

Finding 9 (F6: Post-hoc Isotonic Recalibration Works). *Isotonic regression fitted on 70% of the data brings 11 of 12 configurations below the well-calibrated threshold of 0.05, robustly across 5-fold cross-validation. However, for collapsed models the “correction” merely calibrates a near-constant prediction and does not restore classification quality.*

5.12 Conformal Prediction and Inter-Model Consensus

5.12.1 Conformal Prediction Achieves Target Coverage

5.12.2 Inter-Model Consensus

5.13 Baselines and Composite Utility Score

5.14 Summary of Results

Chapter 6

Conclusion and Future Work

6.1 Summary of Findings

6.2 Answer to the Research Question

6.3 Contributions

6.4 Limitations

6.4.1 External Validation

The present study relies on a single benchmark, TrialGPT-Criterion-Annotations [2], chosen because it is the only public corpus offering per-criterion multi-level expert annotations with an explicit inclusion/exclusion distinction — properties that are essential for fine-grained calibration analysis. External validation on independent datasets was not carried out, for three complementary reasons.

First, existing alternatives are not directly comparable. TREC Clinical Trials 2021/2022 and SIGIR 2016 operate at the trial level rather than the criterion level and use only three ordinal relevance grades, which precludes direct replication of the calibration protocol used here. Adapting the hypotheses to these datasets would not constitute a validation but a distinct study with different endpoints.

Second, the mode collapse phenomenon documented in Section 5.6 appears intrinsically linked to the availability of an evasion label (“not enough information”) in the label space. Datasets that do not include such a label cannot be used to test whether this behavior generalizes: they would trivially prevent it by design.

Third, prospective validation on real clinical data would require ethical approval, hospital partnership, and integration into a live recruitment workflow. These constraints situate external validation as a research project in its own right, complementary to the present work rather than a subordinate step.

External validation nevertheless remains an important direction for future work.

6.4.2 Model Scale

6.4.3 Prompt Space

6.5 Future Work

6.5.1 External Validation on Independent Benchmarks

6.5.2 Multi-Agent Adversarial Debate for Calibration

6.5.3 Investigating the Origin of Mode Collapse

6.5.4 Extension to Larger and Multilingual Models

6.6 Closing Remarks

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Appendix A

Detailed Results Tables

Appendix B

Full Prompt Specifications

B.1 Prompt A — Generic (Baseline)

```
1 Classify whether this patient meets the eligibility criterion.
2
3 Patient: {note}
4 Criterion ({type}): {criterion}
5
6 Choose exactly one label from: included, not included, excluded
7   ,
8   not excluded, not enough information, not applicable.
9
10 Answer :
```

Listing B.1: Prompt A for Instruct models

B.2 Prompt B — Adapted with Expert Guidance

Appendix C

Individual Reliability Diagrams

(a) Mistral-Base, Prompt A

(b) Mistral-Base, Prompt B

Figure C.1: Reliability diagrams for the 12 configurations.

Appendix D

Code and Data Availability

D.1 Repository Structure

D.2 Data Availability

D.3 Reproducing the Experiments

D.4 Computational Requirements

Appendix E

Generative AI Usage Declaration

Tools Used

During the preparation of this thesis, the following generative AI tool was used as an assistant:

- **Claude** (Anthropic), primarily the Sonnet and Opus series model families, accessed via the `claude.ai` interface.

Purpose of Use

Generative AI was used for:

- **Ideation and design discussions:** brainstorming experimental designs, discussing methodological trade-offs, and refining the research question.
- **Code assistance:** writing and debugging Python code for the evaluation pipeline, prompt engineering iterations, and statistical analyses.
- **Writing assistance:** refining the phrasing of prompts, drafting progress reports, and revising English formulations of methodological descriptions.
- **Bibliographic exploration:** identifying candidate references and papers relevant to the state of the art.

Purpose of Non-Use

Generative AI was *not* used for:

- Generating experimental data or results.
- Fabricating statistical outcomes or numerical values.
- Producing scientific claims not backed by our own experiments.

- Writing the final thesis text verbatim (the author remains responsible for all formulations).

Verification and Author Responsibility

All AI-generated content used in this work was:

- Reviewed, verified, and adapted by the author before inclusion.
- Cross-checked against primary sources for any factual claim.
- Understood by the author, who takes full responsibility for the scientific and methodological choices reflected in this thesis.

The scientific reasoning, experimental design, execution, and interpretation of results are the sole responsibility of the author.

Appendix F

Detailed Mode Collapse Analysis

F.1 Prediction Distribution per Configuration

F.2 Simulation: Forcing Decisive Predictions

F.3 Composite Utility Score Ranking

F.4 Cross-Validation Stability

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